



Project Documentation & Technical Dossier

A focused, WASSCE-ready study platform for Ghanaian Senior High School students — with role-based dashboards, real past questions, an AI study tutor, curated national curriculum resources, and an interactive health-education hub.

Product: PhorLearn SHS · **Event:** Hackathon '26 · **Stack:** Next.js 15 · React 19 · TypeScript · MySQL 8 · Tailwind CSS · Google Gemini

Audience: SHS students (15–18), teachers, parents/guardians, administrators

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1. Executive Summary

PhorLearn SHS is a full-stack web application that helps Ghanaian Senior High School students prepare for the **WASSCE** (West African Senior School Certificate Examination). It deliberately narrows its focus to the SHS level so the experience feels serious and exam-driven rather than playful — clean typography, dark accents, and a professional tone an SHS3 student cramming for finals would actually trust.

The platform serves four roles — **Student, Teacher, Parent/Guardian, and Admin** — each with a tailored dashboard. Students get lessons, real past-question papers with timed quizzes and instant feedback, an AI Study Helper, a curated resource library of official NaCCA/WAEC materials, career guidance, and an interactive Health & Wellness hub. Teachers manage classes, set assignments and upload lessons. Parents monitor a ward's performance. Admins see platform-wide statistics.

At a glance: 4 user roles · 8+ subjects · timed WASSCE papers with a closed data-loop into progress tracking · context-aware AI tutoring · curated national-curriculum resources · full keyboard & screen-reader accessibility pass.

WHY THIS EXISTS

2. Problem Statement

SHS students in Ghana face a high-stakes terminal exam (WASSCE) but study with fragmented, low-quality tools. The specific pain points the project targets:

- **Scattered resources.** Past questions, syllabi and notes are spread across random websites, WhatsApp groups and photocopies — hard to trust and hard to find.
- **No instant feedback.** Students practise questions without knowing why an answer is right or wrong, so mistakes are repeated.
- **Limited access to help.** One teacher to many students means individual questions often go unanswered, especially after school hours.
- **Weak progress visibility.** Students, parents and teachers rarely have a shared, data-driven picture of where a learner actually stands.
- **Generic ed-tech.** Most apps are built for younger learners or non-Ghanaian curricula, with playful UIs that don't respect a 15–18 year-old preparing for a national exam.
- **Health literacy gap.** Wellbeing and human-biology knowledge are often taught as dry text, losing students who don't enjoy reading.

3. The Solution

PhorLearn answers each problem with a focused, integrated feature:

PROBLEM	PHORLEARN SOLUTION
Scattered resources	A curated Resource Library of verified NaCCA curriculum, WAEC past papers, teacher manuals and video lessons — searchable and filterable, organised per subject.
No instant feedback	Timed past-question quizzes mark each answer instantly, show the correct option and a plain-language explanation, then a full review screen.
Limited access to help	A context-aware AI Study Helper (Google Gemini) embedded in every lesson, knowing the current subject & topic, available any time and free.
Weak progress visibility	Quizzes feed a closed data loop — updating subject progress, weekly stats, quiz averages and achievements — surfaced on student, parent and teacher dashboards.
Generic ed-tech	An SHS-only , professional UI in real Ghanaian curriculum language (WASSCE, NaCCA, programmes, SHS1–3).
Health literacy gap	An interactive Health & Wellness hub — tap-to-explore body systems, an exercise plan, nutrition with local foods, and a myth-buster game.

4. Product Design Requirements (PDR)

4.1 Scope & tone

SHS-only. No crèche/nursery/primary/JHS flows, no education-level selection, no kid-friendly UI. The tone is a serious study platform: dark accents, clean typography, professional.

4.2 Target users (roles)

ROLE	PRIMARY GOAL
 Student	Learn, practise past questions, get AI help, track progress.
 Teacher	Manage a class, set assignments, upload lessons, view performance.
 Parent / Guardian	Monitor a ward's performance, activity and focus areas.
 Admin	Oversee platform stats, users and content.

4.3 Screen inventory

- Welcome / role selection → Sign up → Log in
- Role dashboards: Student, Teacher, Parent, Admin
- Subject Hub → Lessons (with AI Study Helper)
- Past Questions (paper list → timed quiz → results & review)
- Resource Library · Health & Wellness · Career Guidance · Progress
- Profile (avatar upload) · Help / FAQ
- System: custom 404, error boundary, loading skeleton

4.4 Functional requirements

- **Auth:** email/password sign up & login, role chosen at signup, sessions persisted; protected routes redirect unauthenticated users to login.
- **Student dashboard:** subject progress bars, WASSCE countdown, weekly stats (lessons, study time, quiz average), upcoming homework, quick actions.
- **Past questions:** papers by subject & year; one question at a time; A–D options; instant marking + explanation; score + timer; review; attempt history stored.
- **Data loop:** finishing a quiz recomputes quiz average, updates subject progress, and unlocks achievements.
- **AI Study Helper:** subject/topic-aware system prompt, chat history persisted per topic, loading and error/retry states.
- **Teacher:** class roster with performance bars, class average, assignment list; set assignment & upload lesson forms.
- **Parent:** ward subject performance, weekly activity, focus-area warnings (<65%), email report action.
- **Admin:** platform stats, user breakdown by class level, top subjects table.

4.5 Non-functional requirements

- **Usability:** validated against Nielsen's 10 Heuristics and Shneiderman's 8 Golden Rules.
- **Accessibility:** keyboard focus, ARIA live regions/labels, skip-link, reduced-motion, label associations.
- **Performance:** server components + route-level loading skeletons; small client bundles.

- **Security:** hashed passwords, signed httpOnly session cookies, server-side route guards, server-protected AI key with rate-limiting.
- **Maintainability:** typed data layer, content kept in structured TypeScript datasets.

5. Technology Stack

LAYER	TECHNOLOGY	WHY
Framework	Next.js 15 (App Router)	Server components, server actions, API routes, middleware, file-based routing.
UI library	React 19	Component model; client components for interactive widgets.
Language	TypeScript	Type-safety across data layer, content datasets and components.
Styling	Tailwind CSS 3	Token-driven design system (custom ink/brand/gold/success/danger palette).
Database	MySQL 8.0	Relational store for users, profiles and all app data.
DB driver	mysql2 (promise pool)	Connection pooling + a small Supabase-style query-builder shim.
Auth	jose (JWT, HS256) + bcryptjs	Signed httpOnly session cookies; bcrypt password hashing.
AI	Google Gemini (gemini-2.5-flash)	Free-tier, context-aware study tutoring via a secured server route.
Runtime/tooling	Node.js · npm · ESLint	Build, lint and dev server.

Architectural note: the project originally targeted Supabase (Postgres) and was migrated to **MySQL + custom JWT auth**. A query-builder shim (`lib/db/queryBuilder.ts`) preserves a Supabase-like `.from().select().eq()` API so page code stayed largely unchanged through the migration.

6. System Architecture

6.1 Request & auth flow

1. A visitor signs up (choosing a role) or logs in. The server action hashes/verifies the password with **bcrypt**.
2. On success, a **JWT session** is signed (HS256, via `jose`) and stored in a signed **httpOnly cookie**.
3. **Middleware** verifies the session on every `/dashboard/*` request and redirects unauthenticated users to `/login`.
4. Server components call `requireProfile()` which reads the session and loads the user's profile/role, redirecting if the role doesn't match the page.

6.2 Data layer

A singleton **mysql2 connection pool** (`lib/db/pool.ts`) backs a typed `query()` helper. A lightweight **query-builder shim** mirrors the Supabase API so pages read data declaratively. Per-user demo data is seeded programmatically on signup (`lib/seed.ts`).

6.3 Content layer

Curriculum content lives in structured, typed datasets rather than the database, making it easy to review and extend: `lib/lessons.ts`, `lib/pastQuestions.ts`, `lib/resources.ts`, `lib/careers.ts`, `lib/health.ts`.

6.4 AI integration

The browser never sees the Gemini key. The chat widget calls an internal route (`/api/chat`) that authenticates the user, applies **in-memory rate-limiting**, injects a WASSCE-focused system prompt with the current subject/topic, calls Gemini server-side, and persists the turn to `chat_messages`.

7. Comprehensive App Walkthrough

Every part of the app, what it does, and what you should know about it.

ONBOARDING

7.1 Role selection, Sign up & Log in

[/signup](#) · [/login](#)

Signup is framed as **Step 1**: the user picks a role from four cards, then fills a form. Students additionally choose their **class (SHS1–3)** and **programme** (e.g. General Science). Passwords require a minimum length, a **confirm-password** field with live match-checking, and a show/hide toggle. On success a session is created and the user lands on their role’s dashboard.

Know this: the role chosen at signup determines which dashboard and navigation the user sees for the whole session.

STUDENT · HOME

7.2 Student Dashboard

[/dashboard/student](#)

The student’s control centre, fully data-driven. It shows: a **WASSCE countdown**, “**Continue Studying**” subject-progress bars (each deep-links to that subject’s lessons), **upcoming homework**, a “**This Week**” stats panel (lessons done, study time, average quiz score), and **Quick Actions** to every major area. Empty states are friendly when a new account has no data yet.

LEARNING

7.3 Subject Hub & Lessons

[/dashboard/student/subjects](#) · [/dashboard/student/subjects/\[subject\]](#)

The Subject Hub lists **Core** and **Elective** subjects as cards (English, Core & Elective Maths, Integrated Science, Social Studies, ICT, Biology, Chemistry, Physics). Opening a subject shows a **topic sidebar** and a lesson area with notes and worked examples, a **Study Resources** panel, and resource buttons (Video, Download Notes, Practice Quiz that deep-links to the matching past paper).

AI TUTORING

7.4 AI Study Helper

[embedded on every lesson page](#) · [/api/chat](#)

A side-panel chat that already knows the **subject and topic** being studied, so answers are WASSCE-specific and concise. It shows **suggested starter prompts**, chat bubbles, a typing indicator, graceful error handling with **Retry**, and **persistent history per topic** so each topic keeps its own thread.

Know this: it’s free (Gemini free tier) and the API key is protected server-side.

PRACTICE

7.5 Past Questions & Timed Quiz

[/dashboard/student/past-questions](#) • .../[subject]/[year]

Pick a subject and year (papers tagged *Official paper* vs *Exam-style*, with a legend), plus official download links (NaCCA syllabus, real WAEC PDFs). The quiz presents one question at a time with a **countdown timer**, **progress bar** and running score. Each answer is marked **instantly** with the correct option and an explanation. At the end you get a **score**, **save confirmation** and a **full review**.

Know this: progress is **saved to the browser** (survives refresh), you can move **back to previous questions**, use **keyboard shortcuts** (A–D, ← / →, Enter), and a **leave-page guard** warns before you abandon a live quiz. Submitting recomputes your average and updates subject progress.

REFERENCE

7.6 Resource Library

[/dashboard/student/resources](#)

A curated, verified set of Ghanaian SHS study materials grouped by subject — the official **NaCCA curriculum**, **WAEC past papers**, teacher manuals and video lessons. A **search box** and **type filters** (Curriculum / Past Questions / Textbooks / Videos) make it easy to find a specific resource, with a no-results state.

WELLBEING

7.7 Health & Wellness Hub

[/dashboard/student/health](#)

An interactive, “less-reading, more-discovering” module with five tabs: **Body Systems** (tap a system to reveal its job, key organs, a wow-fact and care tips), **Move & Exercise** (WHO 60-min guidance, exercise groups with reps, a 7-day plan), **Eat Well** (food groups with familiar Ghanaian foods), **Mind & Sleep**, and a **Myth Buster** mini-game with a running score.

GUIDANCE & TRACKING

7.8 Career Guidance & Progress

[/dashboard/student/career](#) • [/dashboard/student/progress](#)

Career Guidance maps subjects to study paths and careers. Progress projects WASSCE grades from quiz performance against a target, helping students see exactly where to push.

ACCOUNT

7.9 Profile & Help

[/dashboard/profile](#) • [/dashboard/help](#)

Profile shows account details and lets any user **upload a profile picture** (stored under `public/uploads/avatars/`, validated for type/size); the avatar then appears in the header. Help is a **role-aware FAQ** reachable from the header on every page.

NAVIGATION

7.10 Screen stepper & shell

Every student screen carries a **Back / Next stepper** (top and bottom) that walks through screens in order with a position indicator, plus a shared header with a skip-to-content link, role badge, Help link and avatar.

7.11 Other role dashboards

[/dashboard/teacher](#) · [/dashboard/parent](#) · [/dashboard/admin](#)

- **Teacher:** class roster with performance bars + class average, assignments with due dates, and working *Set Assignment & Upload Lesson* forms (uploaded materials list below).
- **Parent:** ward subject-performance bars, weekly activity (study time, lessons, streak), focus-area warnings for subjects below 65%, and an Email Report action.
- **Admin:** platform stats (total users, active today, lesson/resource counts), a user breakdown by SHS class level, and a top-subjects-by-sessions table.

8. Data Model (Database Schema)

MySQL 8.0, UTF-8. UUID `char(36)` primary keys, ENUMs for roles/classes, and foreign keys with `ON DELETE CASCADE`. Key tables:

TABLE	PURPOSE
<code>users</code>	Credentials — email + bcrypt password hash.
<code>profiles</code>	Name, role, class, programme, avatar_url (1:1 with users).
<code>subject_progress</code>	Per-subject % progress per student.
<code>weekly_stats</code>	Lessons done, study minutes, quiz average, streak.
<code>homework</code>	Upcoming homework with due dates.
<code>quiz_attempts</code>	Every quiz attempt — subject, year, score, %, duration.
<code>chat_messages</code>	AI Study Helper history, keyed by subject + topic.
<code>achievements</code>	Unlockable badges per student.
<code>classes</code> / <code>class_students</code>	Teacher classes and their roster + scores.
<code>assignments</code> / <code>teacher_lessons</code>	Teacher-set assignments and uploaded lesson materials.
<code>wards</code> / <code>ward_subjects</code>	Parent's linked ward(s) and their subject scores.
<code>exam_config</code>	WASSCE name & date powering the countdown.
<code>platform_stats</code> / <code>top_subjects</code>	Admin dashboard aggregates.

VALIDATION

9. Quality, Usability & Accessibility

The interface was evaluated with two industry usability frameworks, then improved against the findings.

9.1 Nielsen's 10 Heuristics

Strong on visibility of system status, match to the real world, consistency, recognition over recall and aesthetic design. Fixes shipped for the weaker areas: friendly **error & 404 boundaries**, quiz **progress persistence + back-navigation**, **confirm-password**, route-level **loading skeletons**, resource **search**, and a **Help/FAQ** page.

9.2 Shneiderman's 8 Golden Rules

Post-fix the system scores Strong on consistency, informative feedback, dialog closure, error prevention and reduced memory load; Moderate on universal usability and reversibility — which directed the accessibility work below.

9.3 Accessibility pass

- Visible **keyboard focus rings** (`:focus-visible`) and a **skip-to-content** link.
- **ARIA live regions** for quiz feedback, the timer, save status and AI replies; `role="alert"` on form errors.
- Proper **label associations** (htmlFor/id) and autocomplete hints on all auth fields.
- `aria-current` on active nav, `aria-expanded` / `aria-selected` on toggles/tabs, decorative icons hidden from screen readers.
- **Reduced-motion** support via `prefers-reduced-motion`.

SAFETY

10. Security Considerations

- **Passwords** are hashed with bcrypt — never stored in plain text.
- **Sessions** use signed JWTs in httpOnly cookies (not readable by JavaScript).
- **Route protection** via middleware + server-side `requireProfile()` role checks.
- **AI key** is server-only; `/api/chat` requires auth and applies rate-limiting.
- **Uploads** are validated for MIME type and size (≤ 4 MB) before being written.
- *Dev-only note:* `NODE_TLS_REJECT_UNAUTHORIZED=0` is set locally to work behind a corporate SSL proxy and must never ship to production.

RUN IT

11. Setup & Running the App

1. Install **Node.js** and **MySQL 8.0**.
2. In MySQL Workbench, run `phorlearn-app/supabase/schema.sql` to create the `phorlearn` database and tables.
3. Copy `.env.local.example` to `.env.local` and set MySQL credentials, `AUTH_SECRET` and `GEMINI_API_KEY`.
4. From `phorlearn-app/`: `npm install`, then `npm run dev`.
5. Open `http://localhost:3000`, sign up, and explore.

Tip: per-user demo data (subjects, stats, homework, ward/class records) is generated automatically on signup, so a fresh account immediately shows a populated dashboard.

WHAT'S NEXT

12. Future Roadmap

- **Teacher reversibility:** edit/delete assignments & uploaded lessons.
- **Quiz controls:** a “1 minute left” warning before auto-submit and an optional untimed practice mode.
- **Universal usability:** localisation for additional Ghanaian languages and a text-size control.
- **Self-service password reset** and email verification.
- **Embedded video lessons** (verified inline players) alongside the search links.
- **Real assessment analytics** for teachers (per-topic weakness heatmaps).